Investigating the Cognitive Plausibility and Linguistic Interpretability of Large Language Models through Anaphora Resolution

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Introduction

Goal. To assess whether large language models (LLMs) approximate human judgments in anaphora resolution.

Motivation. A central concern for cognitive science is **cognitive plausibility**: To what extent LLMs process and interpret language in ways that align with humans?
We focus on **referential ambiguity in anaphora resolution**:

- arises when multiple discourse entities are available as potential antecedents
- interpretation draws on **syntactic prominence** and **pragmatic bias**

Example. *When Mary called Emma yesterday,* ***she*** *asked about the report.*

- **Antecedents:** Mary (subject), Emma (object)
- **Syntactic bias:** subject $\succ$ object
- **Pragmatic bias:** context can favor either

... didn't answer $\rightarrow$ Emma *... left a voice mail* $\rightarrow$ Mary

Research Questions. *Do transformer-based LLMs integrate syntactic prominence and pragmatic bias in ways similar to humans in their responses? Do self-attention in transformers reflect antecedent–anaphor relationships and show sensitivity to syntactic and pragmatic biases?* (**Case study:** Turkish, contrasting null and overt pronouns.)

Approach.
- Human baseline: offline comprehension task
- LLM analysis: output behavior (prompting) and internal mechanisms (self-attention; antecedent tracking)
- Metrics: mixed-effects logistic regression, Pearson r correlation, entropy

Turkish Anaphora and Prominence

Turkish Subject Anaphora. Turkish is a canonical *pro*-drop language that permits **null subjects**. However, null and overt subject pronouns are not freely interchangeable (Enç, 1986; Taylan, 1986):

- **Null pronouns** are associated with topic continuity and highly accessible antecedents
- **Overt pronouns** occur in marked contexts such as contrast or topic shift

Prominence and anaphoric asymmetry. A central concept in accounts of anaphora resolution is prominence, understood as the relative accessibility or activation of discourse entities in memory.

- Subjects are typically treated as the most prominent antecedents
- More prominent referents are preferentially expressed using less informative anaphoric expressions
- In intra-sentential anaphora (Carminati, 2002):
  - Null pronouns show a strong bias toward subject antecedents
  - Overt pronouns tend to favor non-subject antecedents

Pronoun interpretation draws on multiple sources of information available in the discourse, including subjecthood and pragmatic/contextual biases.

Experiment 1: Humans

Participants. 23 native Turkish speakers (9 male, 14 female; age 18–35).

Design. 93 items, 2 × 3:

- **Anaphor:** null vs. overt
- **Bias:** neutral / subject / object

Neutral	Subject-Biased	Object-Biased
Elif Ayşe'yi gördüğünde, (o) yemek yiyordu.	*Elif Ayşe'yi rezil edince, (o) özür diledi.*	*Elif Ayşe'yi azarlayınca, (o) ağladı.*
'When Elif saw Ayşe, (she) was eating.'	*'When Elif embarrassed Ayşe, (she) apologized.'*	*'When Elif scolded Ayşe, (she) cried.'*

Task. Choose antecedent: *subject / object / someone else*.

Analysis. Mixed-effects logistic regression (anaphor × bias; random intercepts for participants and items).

Results

- Humans show a strong syntactic prominence effect: null pronouns favor subjects ($p < .001$), whereas overt pronouns show reduced subject and increased non-subject interpretations ($p < .001$). Subject bias affects both pronoun types without interaction (all ps $< .01$). In contrast, object bias shows a significant interaction with anaphor type ($\beta = -1.476$, $p < .001$), indicating a weaker effect for overt pronouns. **See Fig. 1 (col. 1) for human results.**

References

Carminati, M. N. (2002). *The processing of Italian subject pronouns*. University of Massachusetts Amherst. **Taylan, E. E. (1986).** Pronominal versus zero representation of anaphora in Turkish. In *Studies in Turkish linguistics*. John Benjamins. **Enç, M. (1986).** *Topic Switching and Pronominal Subject in Turkish*. In D. I. Slobin & K. Zimmer (Eds.), *Studies in Turkish Linguistics* (pp. 195–209). John Benjamins.

Experiment 2: Prompting LLMs

Models. **GPT-4o** (OpenAI; autoregressive chat model); **LLaMA-4**-Scout-17B-16E-Instruct (Meta); Meta-**LLaMA-3**-8B-Instruct (Meta); **Turkish-LLaMA-3**-8B-Instruct-v0.1 (TR-LLaMA-3; Turkish fine-tuned variant) (autoregressive, instruction/chat models).

- GPT-4o was accessed via the OpenAI API; LLaMA-4, LLaMA-3, and TR-LLaMA-3 were accessed via Hugging Face.

Prompting. Prompts mirror the human task:

Prompt	*When Hasan phoned Ayşe, (he/she) was eating.* Who was eating? **Choose one of the options:** Hasan; Ayşe; someone else. **Answer:**
Response	In the given sentence, Ayşe was eating. Therefore, the correct answer is 'Ayşe'.

Table 1. Example prompt and model response. Bolded elements indicate the instruction appended to elicit constrained responses.

Sampling. 100 iterations per item; max response length: 128 tokens; LLaMA-4: temperature = 0.7, top_p = 0.95; LLaMA-3 / TR-LLaMA-3: temperature = 0.6, top_p = 0.9.

Coding. Outputs manually coded: subject / object / someone else; invalid outputs (merged names, no choice) excluded

Analysis. Responses aggregated into sentence-level distributions; identical binomial mixed-effects models (human vs. model); metrics: coefficients (direction & magnitude), Pearson r, entropy.

Results

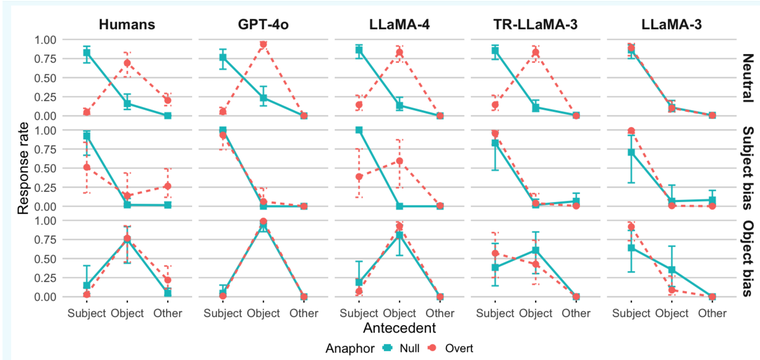


Figure 1. Human and model response rates by bias and anaphoric expression (95% CIs).

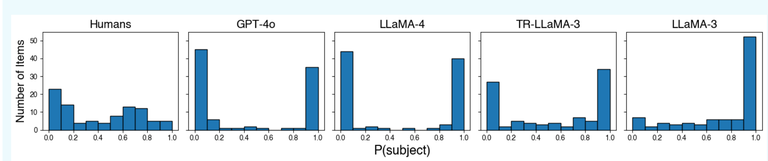
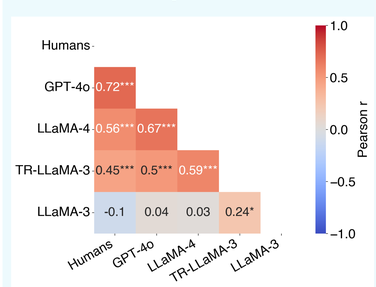


Figure 2. Item-level variation for subject responses in humans and LLMs.



System	Mean Entropy
Humans	0.663
GPT-4o	0.139
LLaMA-4	0.085
TR-LLaMA-3	0.322
LLaMA-3	0.341

Table 2. Mean entropy of subject response rates.

Figure 3. Correlation of subject response rates.

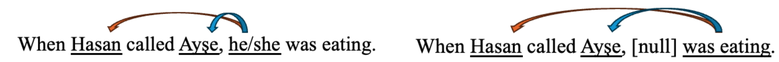
- Models only partially align with human anaphora resolution, differing in **direction** (preferred antecedent) or **magnitude** (strength of the bias).
- Most models reproduce the core **syntactic asymmetry** (null $\rightarrow$ subject, overt $\rightarrow$ non-subject).
- GPT-4o best matches human responses (correlation: $r = 0.72$), but overweights pragmatic bias.
- LLaMA-4 captures the **syntax–pragmatics interaction** (regression), but shows **item-specific variability**, suggesting reliance on lexical cues (e.g., *humiliate–apologize* $\rightarrow$ near-categorical object).
- All models exhibit **lower entropy** than humans, tending toward near-categorical outputs (close to 0 or 1) rather than graded judgments, indicating more deterministic behavior.

Experiment 3: Self-Attention

Model. **LLaMA-4** (decoder-only transformer with masked multi-head self-attention).

Materials. Same 93 stimuli as Exp. 1 and 2; questions and response options removed.

Measure. Attention weights from all layers/heads; relative attention to subject vs. object computed; **source**: overt pronoun or verb; **target**: subject vs. object:



Analysis. Relative subject attention computed per item, layer, and head; correlated with model subject-response rates (Pearson r); FDR correction applied across heads ($q < .05$)

Results

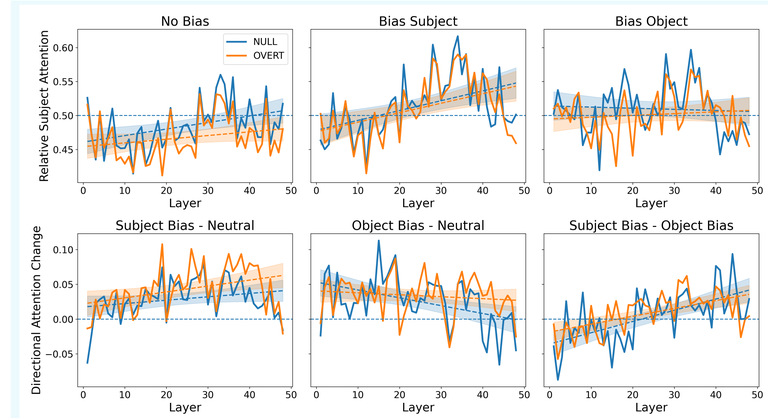


Figure 4. Mean relative subject attention from the verb across layers by bias (row 1); contrasts between bias conditions (row 2).

Source	Anaphor	Neutral	Subject Bias	Object Bias
Verb	Null	0.4844	0.5138	0.5100
	Overt	0.4675	0.5104	0.5013
Overt	Null	–	–	–
	Overt	0.4402	0.4748	0.4699

Table 3. Mean relative subject attention (model-avaraged) by attention source and condition

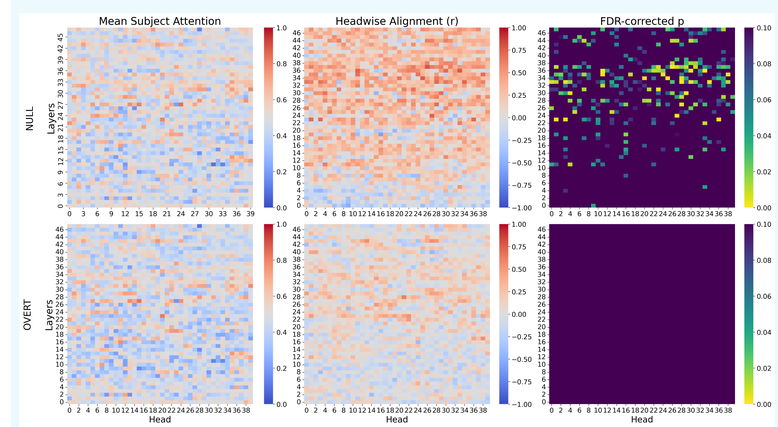


Figure 5. Headwise results for null (top) and overt (bottom) pronoun conditions, with attention computed from the verb. Columns show (left) mean subject attention, (middle) correlations with model subject responses, and (right) FDR-corrected p-values.

- Attention is overall more subject-oriented, with a stronger subject preference for null than overt anaphoric expressions; this difference is reduced in subject-biased contexts.
- Subject attention increases in middle-to-high layers across conditions, with a late-layer reversal under object bias for null pronoun ($p < .05$), but not significantly for overt pronoun.
- A subset of heads in middle-to-high layers shows significant positive correlations with model subject-response rates (FDR-corrected), but this attention–behavior alignment is confined to null anaphoric expressions; overt conditions show no significant correlations after FDR correction.
- Overall, attention provides only limited evidence of encoding antecedent choice, and may reflect structural or positional properties rather than selective referential computation.